

Course: B. Tech.

Branch : AI & DS

Semester : III

Subject Code: BTAIES304

Subject Name: Computer Architecture & Operating Systems

Max Marks: 60

Date: 06/07/2024

Duration: 3 Hr.

Instructions to the Students:

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

	(Level/CO)	Marks
Q. 1 Solve Any Two of the following.		12
A) Explain about address sequencing in control memory with neat diagrams?	L1/CO1	6
B) What are the basic operations performed by the processor?	L1/CO1	6
C) Write briefly about computer fundamental system?	L1/CO1	6
Q.2 Solve Any Two of the following.		12
A) Explain Contiguous Allocation.	L2/CO2	6
B) Describe Implementation of Page Table.	L2/CO2	6
C) Explain role of interrupts in process state transitions.	L2/CO2	6
Q. 3 Solve Any Two of the following.		12
A) Explain Micro-instruction sequencing with it's execution,	L3/CO3	6
B) Describe Multiple processor organization.	L3/CO3	6
C) Explain Cache coherence and the MESI protocol.	L3/CO3	6
Q.4 Solve Any Two of the following.		12
A) Explain System Design and Implementation.	L4/CO4	6
B) Describe Multiple-Processor Scheduling.	L4/CO4	6
C) Write a note Real-Time Scheduling with it's Algorithms.	L4/CO4	6
Q. 5 Solve Any Two of the following.		12
A) Describe methods for handling Deadlocks.	L5/CO5	6
B) Describe solution to Critical Section Problem.	L5/CO5	6
C) Explain Semaphore Implementation.	L5/CO5	6

*** End ***